



# 01. MongoDB Architecture & Shell Fundamentals



## What is MongoDB? (NoSQL Revolution)

MongoDB is an **open-source, document-oriented NoSQL database** designed for modern web applications. Instead of tables and rows, it stores data in **JSON-like flexible documents** with dynamic schemas.



## SQL (Relational) vs NoSQL (Document)

| Feature       | SQL (MySQL, PostgreSQL)            | NoSQL (MongoDB)                                      |
|---------------|------------------------------------|------------------------------------------------------|
| Data Model    | Strict Tables, Rows & Columns      | Flexible JSON/BSON Documents                         |
| Schema        | Rigid Schema (pre-defined columns) | Dynamic Schema (fields can vary across documents)    |
| Relationships | JOINS & Foreign Keys               | Embedded Documents or Object References              |
| Scaling       | Vertical Scaling (larger server)   | Horizontal Scaling (Sharding / distributed clusters) |



## MongoDB Shell (mongosh) Essential Commands

```

mongosh - Terminal
# 1. Start MongoDB Shell
> mongosh
# 2. Display all existing databases
> show dbs
# 3. Switch to or create a database
> use college
# 4. View currently active database
> db
  
```



**LAZY CREATION:** Running `use college` does not create the database on disk until you insert the first document!



## BSON vs JSON Deep Dive & First Document Insert



### Master Comparison: BSON vs JSON (Top Interview Favorite!)

| Feature           | JSON (JavaScript Object Notation)                     | BSON (Binary JSON)                                               |
|-------------------|-------------------------------------------------------|------------------------------------------------------------------|
| Format            | Plain text (UTF-8 string)                             | Binary-encoded format                                            |
| Readability       | Human-readable (easy to inspect)                      | Machine-readable (raw binary bytes)                              |
| Data Types        | Limited: String, Number, Boolean, Array, Object, null | Rich: ObjectId, Date, BinData, Int32, Int64, Decimal128, Regex   |
| Speed & Traversal | Slow parsing (requires scanning entire text)          | Fast traversal (stores length prefixes for rapid field skipping) |
| Storage Overhead  | Text representation incurs space overhead             | Optimized for fast indexing and disk storage efficiency          |



### Anatomy of \_id: ObjectId("65c8a1f2...")

MongoDB automatically assigns a 12-byte globally unique identifier to every document:



### Inserting First Document (insertOne)

```

mongosh - insertOne

// Insert document into student collection
db.student.insertOne({
  name: "Rahul Sharma",
  age: 20,
  marks: 94,
  city: "Delhi"
});
// Return Object: { acknowledged: true, insertedId: ObjectId("65c8a1f2...") }
// Now 'show collections' will display 'student'!
  
```



**EXAM ALERT:** Because BSON stores explicit field lengths, the engine can skip directly to relevant fields without parsing the entire document text!